

BASIC ELEMENTS OF AN INTERNATIONAL TERRESTRIAL REPLY FOLLOWING THE DETECTION OF A SIGNAL FROM EXTRATERRESTRIAL INTELLIGENCE

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Abstract—One of the most tantalizing questions human intelligence has posed itself since the beginnings of time is whether we are alone in the Universe. And if not, where are "the others," and is it possible that these "others" may contact us, with or without our knowledge? Scientific literature abounds with theories pro and con, and if any subject ever lent itself to make-believe it is this one. Connected with the scientific theories on the subject, which are so often misunderstood, popular belief has led to a steady stream of publications which have done nothing if they have not greatly hampered the real issue.

In legal circles the subject of possible contacts with extraterrestrial intelligence was first discussed in the literature in the early 1960's when, among others, Andrew Haley wrote, in 1965, about "Space Law and Government." This publication appeared in the midst of the United Nations preparations for what was to become, in 1967, the Outer Space Treaty. Though the idea of an international protocol governing actions after the detection of a signal from extraterrestrial intelligence has never been on the agenda of the United Nations Committee on the Peaceful Uses of Outer Space, space lawyers continue to think and to write about that subject.

Recent advances in spaceflight, in particular the Voyager missions to the outer planets of our solar system, give the subject a new impetus. The present paper is intended to dwell upon present scientific thoughts about the likelihood that a signal from extraterrestrial intelligence will be detected in the near future. Although this likelihood appears to be remote, it seems, nevertheless, worthwhile that discussion continues in space legal circles regarding the formulation of an international protocol for activities following the detection of a signal from extraterrestrial intelligence. Since the confirmed detection of a signal from extraterrestrial intelligence could well be of crucial importance for the continuing existence of the Earth and its inhabitants, such a protocol cannot be formulated better than at present, when no such detection has been made as yet. Before a protocol can be formulated, however, a great many questions must be answered at least tentatively, e.g. the following ones:

- What are the astronomical, exobiological and philosophical probabilities that a signal from extraterrestrial intelligence will be detected and recognized as such?
- Which person(s) will, in all likelihood, be the one(s) to detect such a signal?
- What should be the criteria to identify an extraterrestrial signal as intelligent?
- To which person(s) should the establishment and application of such criteria be entrusted?
- If a signal is identified beyond doubt as intelligent, which person(s) should undertake to decipher its contents and to establish whether the signal was accidentally received on Earth, or premeditatedly sent to Earth?
- Which person(s) should determine the distance from Earth of the transmitter of the signal?
- Depending on the answers to the latter two questions: what actions should be taken, and by whom?

The present paper offers some replies to these, and other related questions. Following the resume of the replies, the main conclusions of the paper regarding an outline of the proposed protocol are that:

1. The scope of the protocol must be international.
2. An international forum, or committee, representing the entire world population in this matter should be established.
3. The competence and authority of the forum, or committee, should include the opportunity to decide that no activities following the detection of a signal from extraterrestrial intelligence shall be undertaken.
4. If the forum, or committee, decides that a return signal shall be sent, the forum or committee should have the authority to decide upon the contents as well as upon the means of transmission of the return signal.

1. INTRODUCTION

One of the most tantalizing questions human intelligence has posed itself since the beginning of time is whether we are alone in the Universe. And if not, where are "the others," and is it possible that these "others" could contact us, with or without our knowledge?

The investigations leading to a possible answer to these questions can be divided in two parts, the one preceding the other: a scientific part and a philosophical one. The order of the division is based on the assumption that the first contact with ETI will, most probably, take place through advanced scientific instruments. In our opinion, it is only after a possible contact with

ETI has been established that philosophical, social and legal implications may play a role.

It seems prudent, however, to start thinking about the impact of a possible contact with ETI and the procedures to be developed for such contacts in an early stage. For the present paper, this means that, in the first part, the status of scientific research will be reviewed, especially regarding the results from astronomical, astrophysical and exobiological probabilities of terrestrial contacts with ETI. The outcome of the scientific review will influence to a large extent the contents of the procedures to be developed for the terrestrial handling of the contacts with ETI. Out of the many possible procedures in many possible

fields of terrestrial society which will be deeply influenced once a contact with ETI has been established beyond any doubt, we shall concentrate on the basic elements of an international protocol that, in turn, may contribute to enhancing international law-making. Under this protocol, also the questions as to the contents of a message shall be considered.

2. THE STATUS OF SCIENTIFIC RESEARCH REGARDING ETI

The first question is: what is "intelligence"? So far as known, no generally accepted definition of "intelligence" does exist. One proposal encompassing many, but by no means all, possible definitions of "intelligence" refers to "the quality of a process, namely the capacity to understand[1]." We accept that definition for the purpose of this paper.

The next question is how to define "contact." Etymologically, the word "contact" derives from the Latin verb *contingere*: the touching of two individual data, be they technical, or human, or animal, or otherwise. In the matter under investigation, two types of such contacts are feasible: from ETI to the Earth, and vice versa. There is no evidence that ETI indeed has ever contacted the Earth; if ETI did, however, this has passed unnoticed. A number of different categories of such evidence can be envisaged:

- a. Electromagnetic radiation received from a non-human source has been identified beyond reasonable doubt as a non-natural, i.e. artificial signal;
- b. One or more objects are discovered, either on Earth itself or in its immediate neighborhood, which are identified beyond reasonable doubt as being both artificial and non-human, and which contain no living creatures;
- c. The Earth is visited by extraterrestrial spaceships with living beings on board;
- d. The Earth receives unmistakable and confirmed evidence of the presence on or near it of an alien intelligence that manifests itself in presently unknown and unforeseeable ways.

For each of these categories, a number of different situations in which the relevant evidence is established can be envisaged:

- a.1. The source of the signal evidently is at a distance from the Earth which is at least as great as that of the stars in the solar neighborhood, i.e. of the order of 10 light-years (10^{13} km) or (perhaps much) more. In

that case, a group of competent individuals should be given the tasks, first of all to verify the artificial character of the radiation beyond any reasonable doubt, and next to decode the signal contained in the radiation and to decipher its contents. If, as a result, it becomes clear that the sender was aware of the existence of intelligent beings on Earth and that the message was specifically intended to be received by such (premeditated contact), then a competent international body should decide upon the course of a reaction. It should be borne in mind that the message has been sent a number of years before its reception which is equal to the source distance expressed in light-years, and that the transmission time for a reply message is equally long. Thus, for a source distance of 10 light-years, it will take at least 20 years before a reply to a message can be expected. If, on the other hand, no indication of an awareness of our presence on the part of the sender is found (stray contact), there is no urgent need for a reply. Still it would be wise to discuss the impact of this discovery upon life on Earth in international bodies as mentioned above;

- a.2. The source of the signal is located somewhere near or within our solar system. In that case, it seems most likely that this source is a non-human artifact (spacecraft) and that, consequently, we are in the situation described in category "b" or "c";
- b. If an artificial object has been discovered within our solar system, either by direct verifiable observation or by identifying it as the source of a message, all possible efforts should be made to find out whether it contains a crew of living creatures or not. In case it does, we are in the situation of category "c." If there is no evidence whatever that the object contains living creatures, efforts should be made by a competent body to find out whether the object arrived at the solar system on purpose or by accident. Depending on the outcome of these investigations, appropriate steps should be taken as mentioned in "a" above;
- c. If the solar system in general or the Earth in particular is visited by extraterrestrial spaceships and if it is evident that living intelligent creatures are on board these spaceships, it seems obvious that it is of vital importance that the behavior of the ships and their inhabitants be closely monitored. Unless it is unmistakably clear that the spaceships are on a hostile mission, every

possibility of a hostile action on the part of the Earth should be carefully avoided. Any contact that might be sought by the ship crew should be considered by a competent and responsible international body, lest the visitors should get too unfavorable an impression of the development of intelligent life on Earth;

- d. In this case the only possibility seems to be to react as seems fit, in the spirit as indicated under "c."

It seems evident that the discovery of extraterrestrial, intelligent life will be of great if not vital importance for the continuation of life on Earth. If contact is established only by way of radio signals, the influence may be slow. If, on the other hand, contact is direct, with visitors close at hand, then everything will depend on the comparative levels of technological knowledge and spiritual wisdom.

The most obvious way to look for evidence of the existence of ETI is by way of radio-astronomical research projects. Since 1960, terrestrial astronomy has been analyzing emissions from cosmic radio sources in order to identify any artificial, intelligent signal that might be present in them. At first, this was done with relatively modest technological means and on an incidental basis, but later on increasingly sophisticated instruments were made available, some of them even on a continuous basis. As of mid-1984, some 50 research projects had been carried out in the USA, the Soviet Union, Australia, Canada, France, Germany and the Netherlands, spending between them some 120,000 hours of observation time[2,3]. Most of these searches were conducted at frequencies between 1400 and 1700 MHz, corresponding to wavelengths between 21 and 18 cm, the so-called "water hole" (1420 MHz or 21.1 cm is a characteristic emission line of atomic hydrogen, and 1667 MHz or 18.0 cm is a line of the OH radical, the two constituents of water). In this spectral region and up to around 100 GHz, the omnipresent cosmic background radiation is relatively low, so this region is suitable for interstellar communication.

Some of the more recent projects are SENTINEL of Harvard University, which began in March 1983, and its successor META, which started in September 1985. These projects are fully dedicated to SETI. At the University of California at Berkeley, a project for shared or parasitic research of radioastronomical data obtained for other purposes, called SERENDIP, was started in 1980. A few years ago it was expanded and is now called SERENDIP II. At

NASA, a SETI program is in preparation which can be compared to META, although it is more flexible.

So far, none of these projects have produced any positive result. The conclusion must be that, scientifically speaking, there is no evidence that ETI has contacted the Earth or that terrestrial science has contacted ETI. Should, however, a contact with ETI be established, this will most probably be through technical devices.

3. THE STATUS OF LEGAL RESEARCH REGARDING ETI

Legal research - in the strict interpretation - cannot yet have been done in view of the consideration that *ius ex facto oritur* - law follows the fact(s). The discussion in legal circles has, therefore, concentrated on philosophical issues of a general nature, such as:

- Will the basics of terrestrial law suffice for the regulation of the changes in terrestrial society which will occur as a consequence of a contact with ETI?
- Depending on the physical characteristics - if any - of ETI and of the place in the universe where extraterrestrial science reveals it to be present: will terrestrial law be entitled to offer its Earth-bound qualifications as a means of coexistence with ETI, or even to enforce it? Haley argues in this respect that "no concept of common law, civil or criminal, which in any matter whatsoever is framed for the purpose of being enforced - except, perhaps, on other human beings - should be projected for the government of intelligent beings who dwell elsewhere in the cosmos[4]."

The utmost prudence in applying standards of terrestrial law in contacts with ETI seems indicated. Up to the present, it is entirely unknown to us if the ETI will be a higher form of intelligence than we are, an equal one, or a lower one. In view of the technological capabilities required for even the basic forms of terrestrial astronomical and astrophysical space research it may be assumed that ETI seeking contact with Earth will be of a higher form of intelligence than the present terrestrial one[5].

Assuming also that the higher the intelligence, the deeper will be the civilization within which it is embedded, there is every reason to suppose that our terrestrial legal system will, in this respect, be of no avail. Law on Earth is, essentially, a means to effectuate justice. It seems self-evident, in any form of intelligence higher than the terrestrial one, the means to

effectuate justice will have been accomplished to a higher degree too.

At present, terrestrial legal systems are, however, the only ones we can deal with, as all future regulations upon possible contact with ETI still have to be hypothetical. In view of the international consequences of a contact with ETI, international law would seem the obvious branch of law to provide guidelines - even if the contact should, as it most obviously will - be established by a national technical device such as a space telescope.

Space law, a branch of international law, has only occasionally concerned itself with the question of guidelines regarding contacts with ETI. The existing space legal regulations in this respect, the Outer Space Treaty 1967 (OST67) and the Moon Agreement 1984 (MA84), have been formulated in too generalized wording to refer to in this matter. There are a few instances in the space treaties which seem applicable in this respect, however, e.g.:

1. The Outer Space Treaty 1967:

Art. I, regarding the intention to carry out the exploration and use of outer space in this case: the search for contacts with ETI from the Earth "for the benefit and in the interests of all countries....."

Art. IX, regarding the principles of cooperation and mutual assistance (between States) and of such cooperation in avoiding "harmful contamination and also adverse changes in the environment....."

Art. XI, regarding the principle that "States Parties to the Treaty.....agree to inform the Secretary General of the United Nations as well as the public and the international scientific community....."

2. The Moon Agreement 1984:

Art. 5 par. 3, regarding the obligation of States Parties to the Agreement to inform "the Secretary General, as well as the public and the international scientific community.....of any indication of organic life (in outer space, or on the moon)."

Also, and this is an interesting new development, the Scientific and Technical Sub-Committee of the United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) has begun in its February 1986[6] session the consideration of two new items on its agenda:

- matters relating to planetary exploration, and
- matters relating to astronomy[7].

It may very well be that a new dimension will be added to space law if the UNCOPUOS Legal Sub-Committee would decide to also put these

items on its agenda in the coming years. At present, international legal specialists try to formulate the basics of necessary rules for a protocol regarding activities once a signal of ETI has been detected.

4. OUTLINE OF THE BASIC ELEMENTS OF AN INTERNATIONAL PROTOCOL

As a result of the foregoing considerations, the following basic elements of a protocol emerge:

1. The protocol must be adaptable to at least the four major present possibilities as sketched hereabove in sections "a" through "d";
2. The protocol should be international in scope;
3. The protocol should describe the form in which a group of competent individuals shall be chosen; that is to say, individuals who are:
 - a. competent to verify the artificial character of the radiation, or any other means of contact of ETI with Earth, beyond any reasonable doubt;
 - b. In view of the character of ETI this group will have to consist of scientists (group A) competent to decode the signal contained in the radiation, or any other means, and to decipher its message.
4. The protocol should provide guidelines for the formation of a second group of individuals, who shall represent, solely for the event of a possible contact with ETI, the entire Earth (group B). The main criterion for the choice of the members of this group has to be the degree of the spiritual wisdom of its members;
5. Group B should:
 - Refrain from all actions and reactions towards ETI until the members of group A have, unanimously, come to the decision that the signal, or any other means of contact, is indeed coming from ETI;
 - Have the authority to decide whether a reaction shall be given, or not;
 - Discuss, in case of a positive decision as to a reaction to be sent, the form and the contents of the reaction with group A;
 - Leave it to group A to actually react;
6. The members of the combined groups A and B (group C) and only if no single member of group C is absent, shall not be chosen according to their nationality, political, economical, or other such qualities; they shall be chosen solely on the basis of being

the world's most gifted in spiritual wisdom, in such a way that they shall have the fullest possible confidence of the international community. Such choice seems to offer the best guarantee that any nationalistic, political or other unfavorable tendency in a return message is avoided;

7. Group C, with all its members present, shall be the sole authority to decide if, and if so, in what form and with what contents, the evidence of contact with ETI shall be released, to whom, and when. As to the release of the evidence of contact with ETI, we do not share the view as expressed by several authors[8] that such evidence should be immediately released and to everyone. The proceedings around the Chernobyl disaster of April have convincingly shown that even the most prudent, stylized and scarce information has led to panic. If the real facts[9,10] would have been released immediately after the event, the behavior of the general public might well have entirely run out of control. In case of evidence of contact of ETI with the Earth, the contact will in all probability influence terrestrial human behavior to a degree that is scarcely imaginable, or comparable to the influence of the Chernobyl event. Should group C, nevertheless, decide on the release of information once ETI has contacted the Earth, then the information must consist of the barest essentials and be offered gradually;
8. Should contact of ETI with Earth have been established by means of the arrival on Earth of terrestrial space ships with living intelligent creatures on board, then the task would fall to group C to initiate, or not, the first contact, guided by such considerations as mentioned under 1.c. hereabove.

5. BASIC ELEMENTS OF AN INTERNATIONAL TERRESTRIAL REPLY

The members of Group C, as established under the Protocol as proposed hereinbefore, will first have to decide the relevance of an international terrestrial reply. That decision will most probably depend on scientific data as well as on the way in which the contact has been established.

Once the decision has been taken that a reply is relevant, it will depend on the character of the message contained in the signal from ETI how group C will formulate the message in the return signal. The message contents of the

terrestrial reply should in our opinion be basic; they should explain:

- that intelligent life is present on Earth;
- that Earth has received the signal and understood its message;
- the kind of organisms existing at present on Earth, especially the kind of organism that formulated the present message: human beings;
- the exact location and size of the Earth in the Universe;
- the fact that the senders of the signal from Earth send it on behalf of the entire Earth;
- the main physical features of human beings, and their number;
- the express desire of the Earth to establish and to maintain friendly relations with the senders of the ETI signal;
- the wish to receive more signals from ETI.

The above elements of a return signal would form an adequate reply to all or most of the categories described hereabove under par. 1, items a. through d. Once a signal from ETI has been received, decoded and replied to, group C should begin to prepare a second reply to a second signal from ETI, if that ever would come, taking into account the distance of ETI and the corresponding travel time of the signals.

6. CONCLUSION

In spite of 27 years of continuous scientific research there is no evidence that we on Earth have ever succeeded in establishing a contact with ETI. There is no scientific evidence either that ETI has ever established contact with the Earth. Scientific research in this respect continues, so either possibility of contact still exists.

At present, international law and especially space law cannot be and is not equipped for the consequences of such contact. In view of the uprooting social effects of contact with ETI, an International Protocol to enter into force upon such contact seems necessary. This Protocol might serve as a means to facilitate the formulation of rules of international law, or space law, in this respect.

The present paper has suggested eight basic elements for such an International Protocol:

- It must be international in scope;
- It must cover, now or upon evidence of contact with ETI, the four main scientific possibilities of such contact;
- It must provide guidelines for the formation of two groups of competent individuals:

- one group consisting of the world's best scientists (group A), and
- one group consisting of the world's most gifted in spiritual wisdom (group B);
- Group A and B together, called group C, will ultimately and authoritatively decide if, and if so, which, action or reaction shall be taken or given upon evidence of contact with ETI;
- Especially for the formation of group B, specific guidelines have been suggested in this paper;
- Group C shall, under all circumstances, be the sole representative of the Earth upon contact with ETI;
- Group C shall be the sole authority to decide whether or not, and if so, in what form and to whom, information shall be released regarding contact with ETI.

Once group C has decided that a return signal from Earth to ETI is relevant, it should formulate the message in the return signal. As to the contents of the message in the return signal, we suggest nine basic elements, all of them containing elementary information about the Earth, with emphasis on the desire of the Earth to establish and to maintain friendly relations with ETI.

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